Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

RoomDrop

- Team name: **Triple-8**
- Product: **RoomDrop**
- Team leader name: **Pete** (team: Pete & AJ)
- Problem Statement: Travellers book hotels early but fear missing a cheaper rate later. RoomDrop lets them book early and still catch the drop. Peace of Mind Guaranteed!

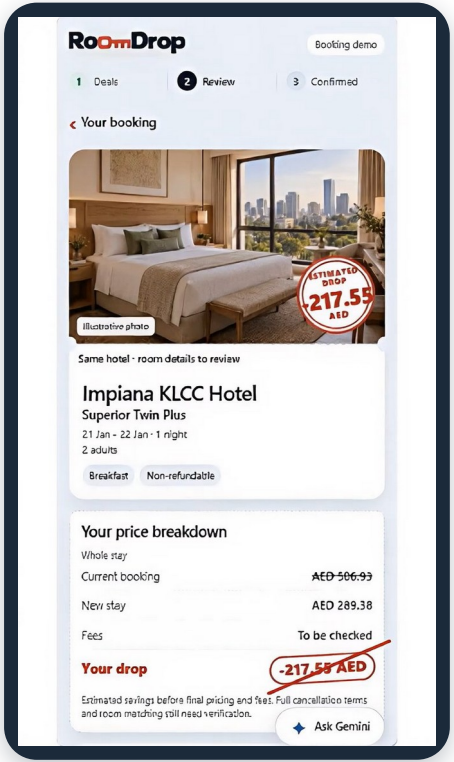
RoomDrop in 10 seconds

RoomDrop

Book early.
Still catch the drop.

A 10-second tour of the live prototype:
find a drop, peek, ask Gemini, confirm.

Full 3-4 min demo: link on the final slide



BEST DROP
-217.55
AED

Real Altovo quotation
Impiana KLCC Hotel, Kuala Lumpur
AED 506.93 to **AED 289.38**

Brief about the idea

RoomDrop

Book early. Still catch the drop.

- Book your hotel early. RoomDrop keeps re-shopping the same stay for you.
- A cheaper rate appears? See the drop, the saving and any flexibility you give up.
- Ask Gemini, tick the trade-off, confirm in one tap.

Built on Google Cloud

Cloud Run
Public app, private agency console and the price-watch job

Gemini on Vertex AI
Gemini 3.8 Flash drop adviser: structured JSON, validated by the server

Cloud Firestore
Bookings, drops, Drop Watch consent and sessions, with transactions and TTL

Opportunities

RoomDrop

How different is it from any of the other existing ideas?

- Alerts only notify; RoomDrop shows a real drop with the saving and the flexibility you give up
- Re-shops the exact stay you booked, no manual cancel-and-rebook
- Runs on the agency's own quotation and supplier feed, not another meta-search

How will it be able to solve the problem?

- Starts from your current booking: same hotel, dates and guests
- Ranks three drops by estimated saving and flags non-refundable risk
- Gemini answers "should I switch?" using only the facts on screen

USP of the proposed solution

- Book early, never overpay: every saving shown as a stamped drop
- Trustworthy AI: Gemini cannot book, refund or invent prices
- One engine, two markets: traveller savings and agency margin

B2C Travellers lock in early and still catch the drop

B2B Agencies re-shop held bookings to cut cost and raise margin

List of features offered by the solution

RoomDrop

LIVE

Booking + 3 drops

Current booking ticket above three ranked drops

LIVE

Saving stamps

Estimated drop per stay, with cancellation and refund warnings

LIVE

Peek at a drop

Price breakdown, room, board and cancellation flag

LIVE

Ask Gemini

Compare all drops or ask about one, in plain language

LIVE

Grounded AI

Replies validated against the booking and the suggested drops

LIVE

Consent + receipt

Explicit tick for lost flexibility, then a selection-only receipt

SOON

Agency console

Private, read-only cases behind Google IAP

NEXT

Drop Watch

Consent, minimum saving and scheduled re-checks with alerts

Process flow diagram or Use-case diagram

RoomDrop

Traveller journey B2C, live today

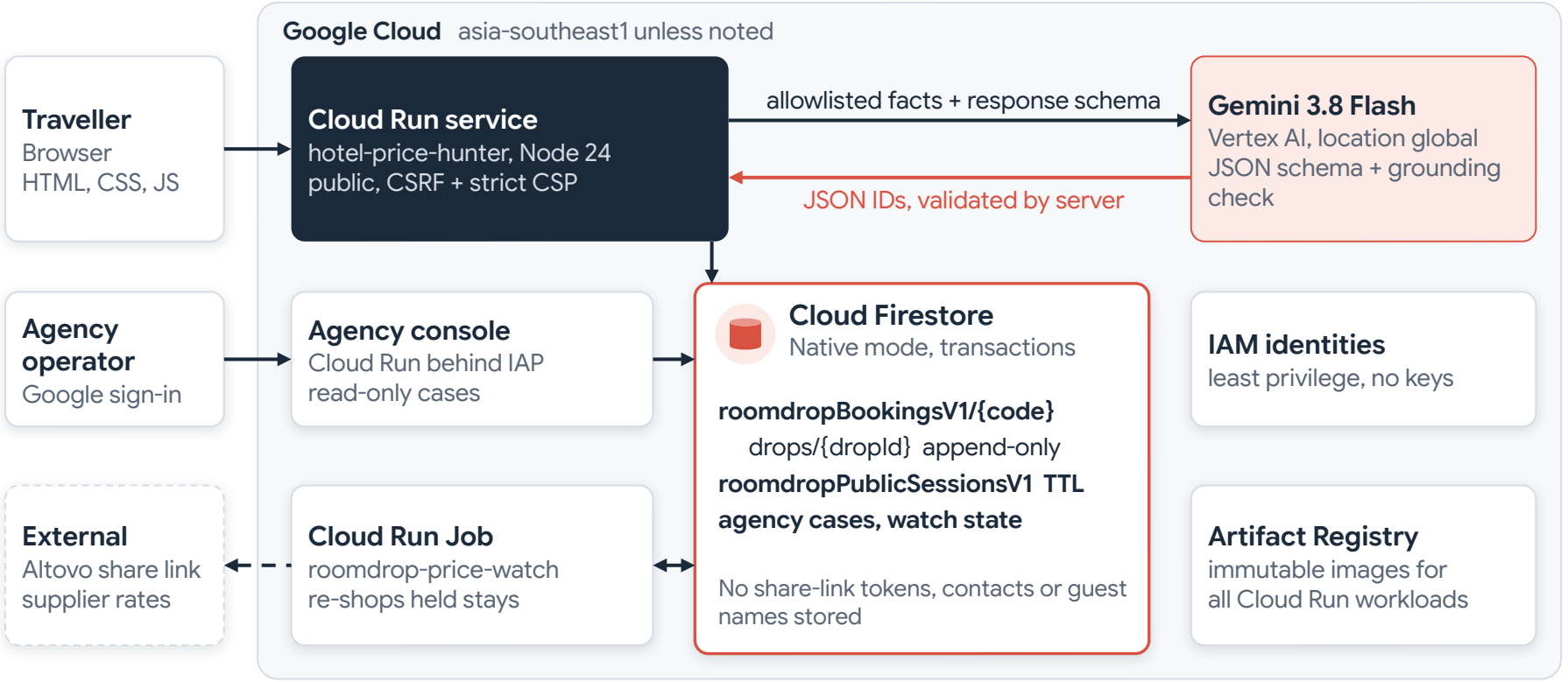
1. **Your booking** — Hotel, dates, guests and price
2. **Find drops** — 3 ranked drops with saving stamps
3. **Peek at a drop** — Price breakdown and room facts
4. **Ask Gemini** — Grounded advice: price vs flexibility
5. **Review** — Tick any lost flexibility
6. **Confirm** — Receipt with the estimated drop

Agency use FUTURE B2B, same engine

- **Share booking** — Agent sends an Altovo booking link
- **Re-shop the stay** — Supplier gateway, same hotel and dates
- **Record drops** — Booking and drops kept in Firestore
- **Review cases** — Private console behind Google IAP
- **Rebook cheaper** — Cut supplier cost, keep margin (next)

Architecture diagram of the proposed solution

RoomDrop



Technologies to be used in the solution

RoomDrop

Frontend

- Vanilla HTML / CSS / JS
- ES modules, no build step
- Strict CSP

Backend

- Node.js 24
- node:24-bookworm-slim
- built-in node:http
- npm 11.6.2

AI

- Gemini 3.8 Flash
- Vertex AI (global)
- @google/genai 2.28.0
- JSON response schema

Data

- Cloud Firestore
- Firestore SDK 9.3.1
- Firebase emulator for local Firestore tests

Cloud / DevOps

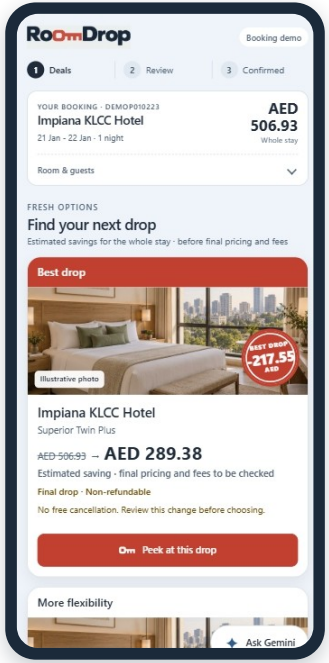
- Cloud Run services + Job
- Artifact Registry
- IAM + Google IAP
- Docker

Testing

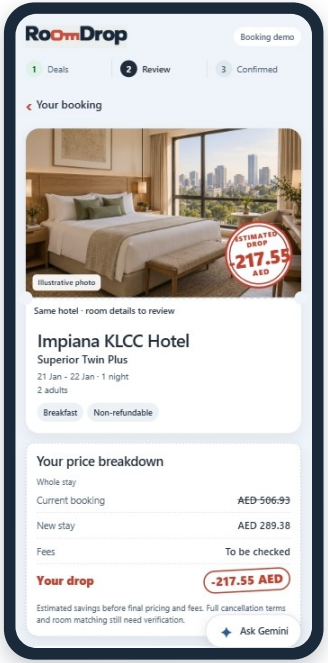
- node:test, 292 pass
- Firestore emulator tests
- Playwright, 320-1440 px

Snapshots of the prototype

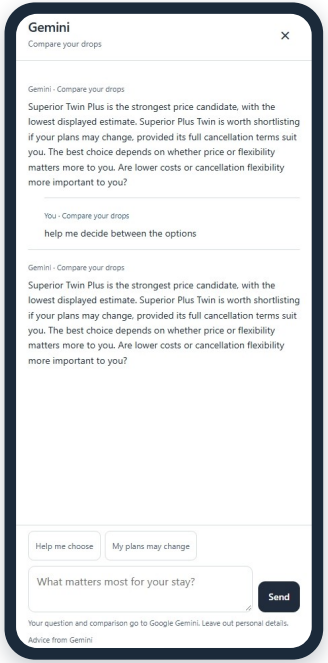
RoOmDrop



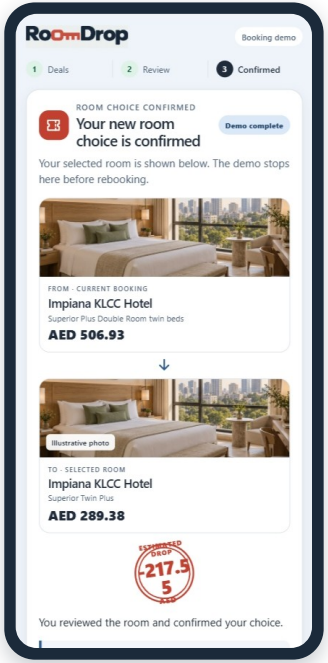
1 Find your next drop



2 Peek at a drop



3 Ask Gemini



4 Confirmed

Prototype Performance report/Benchmarking

RoomDrop

100/100
Accessibility and Best Practices, mobile and desktop

3.3 s
median Gemini reply; 7/7 passed grounding checks

77 ms
warm page TTFB from Malaysia (server ~3 ms)

292 pass
0 failures; 6 skips need the Firestore emulator

Lighthouse (median of 3)	Mobile	Desktop
Performance	75	89
Accessibility / Best Practices	100 / 100	100 / 100
SEO	91	91
First Contentful Paint	1.5 s	0.43 s
Largest Contentful Paint	12.2 s	2.2 s
Total Blocking Time	12 ms	0 ms
Cumulative Layout Shift	0	0.003

How we measured

18 Oct 2026, live Cloud Run URL (asia-southeast1, scales to zero). Lighthouse 12.8.2 x3 per preset; curl x10 sequential from Malaysia; 7 sequential Gemini chat calls.

Next: compress images

Mobile LCP is 12.2 s from 2.6 MB of PNG/JPG. WebP/AVIF and responsive sizes save ~1.2 MB.

Additional Details/Future Development (if any)

RoomDrop

1 Share-link intake

Agent sends an Altovo link; booking and drops persist in Firestore

2 Scheduled Drop Watch

Cloud Scheduler runs the price-watch job daily on live supplier rates

3 Drop alerts

Notify when a drop beats the traveller's minimum saving

4 Real rebooking

Turn a chosen drop into a new booking, cancel the old one, refund

5 Traveller sign-in

Invitations and identity so travellers can return to their drops

6 Agency actions

Act on cases from the IAP console; October UX review fixes

Trust guardrails, live

- Gemini replies are schema-bound IDs, re-validated by the server
- Money is computed by code, never by the model
- Only allowlisted facts reach Vertex AI; no guest data
- Selection only: no booking, payment or refund routes
- Non-refundable drops need an explicit tick
- No service-account keys; agency behind IAP

RoomDrop

Provide links to your:

GitHub Public Repository	https://github.com/pohlipit23/hotel-price-hunter
Demo Video Link (3-4 Minutes)	https://youtube.com/shorts/zKe3UFlgUNc
Final Product Link	https://hotel-price-hunter-2qbcpylngq-as.a.run.app/

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Thank you

